Brief for Mr Abhishek Pitale — From Labour Care Documentation to Maternal Death Review

Seven-day assignment · Nov 2, 2026; Prepared By: Dr Pradeep Atter

Welcome and purpose

Over seven days, we would like you to explore — and demonstrate in a working prototype — whether information recorded while a woman is in labour can be safely reused when her death later has to be reviewed, so that a doctor does not have to reconstruct it.

The assignment tests how you think as a technical leader:

- how quickly you understand two unfamiliar clinical workflows and the relationship between them.
- whether you can see what overlaps, what does not, and what only *looks* as if it overlaps.
- how you design for exhausted users under pressure.
- how you protect the accuracy, traceability and integrity of records that may later be examined closely.

The problem: from labour ward to death review

Doctors and nurses document clinical information during labour. If a maternal death occurs, they must complete an eight-page Facility-Based Maternal Death Review (FBMDR) form within 24 hours.

Maternal death is defined by the WHO as the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration or site of pregnancy, from any cause related to or aggravated by the pregnancy or its management, but not from accidental or incidental causes.

The FBMDR form requires patient demographics, obstetric history, referral details, labour and delivery records, interventions, complications, and circumstances surrounding the death. Much of this information has already been recorded in clinical documentation, including the Labour Care Guide (LCG).

Problem: Clinical information documented during labour is not automatically transferred to the FBMDR form. Healthcare workers must manually retrieve, verify, and re-enter the same information, resulting in duplicated work, increased documentation burden, and potential transcription errors.

Workflow A: labour documentation and the WHO Labour Care Guide: Provided as PDF

Workflow B: facility-based maternal death review: Provided as PDF

Design principles: provenance, integrity and clinical honesty

> ***Capture once. Validate appropriately. Reuse where permitted. Report accurately.***

Ensure the following:

1. **Every imported value is traceable.**
2. **The original is never changed by the review.**
3. **Authorship is preserved.**
4. **Contemporaneous and retrospective are kept apart.** Information recorded at the time is different from information added after the death. Entries made to the labour record after the time of death should be clearly identifiable.
5. **Clinician's explicit decisions are respected.**
6. **Reuse is offered, not imposed.** The clinician sees what is proposed for reuse, accepts or rejects it, and is accountable for what goes into the review.
7. **Nothing is silently changed after submission.** Corrections are visible, attributed amendments.
8. **Reviews are confidential and for learning.** ***Design language and permissions should support honest reporting, not blame.***

Who you are designing for. A resident at 4 a.m., post-call, on phone, who has just been part of a failed resuscitation and still has other patients. They will be interrupted. They need to see at a glance what the labour record already supplies, what is still missing, and what does not apply — and to trust that what is reported is what was recorded.

What we want you to build: the demo journey

A working application, on fictional data, that holds labour records, lets a doctor start a death review from a woman's record, proposes what can be reused with its source visible, guides the doctor through everything else, and produces a structured review output that an FNO can approve — with nothing typed twice and nothing claimed that was not recorded.

The journey we expect to see in your demo

1. **Labour is documented.** A fictional woman's labour record exists, with observations entered over time by more than one person (for example a nurse and a resident), including at least one value crossing an alert threshold and at least one gap.
2. **She dies.** The time of death is recorded.
3. **A review begins.** A resident opens a new death review linked to her record.
4. **Reuse is proposed.** The system shows which review items it can fill from the labour record, how, and from which entries — and which it cannot.
5. **The resident decides.** They accept, reject or correct each proposed value; corrections do not alter the labour record.
6. **The rest is completed.** Everything the labour record cannot supply is captured through a guided, conditional experience, resumable after interruption.

7. **Judgements are the resident's.** Cause of death, contributing factors and the narrative are entered as the clinician's own conclusions, with relevant labour observations shown alongside as evidence.
8. **Review and approval.** The resident submits; an FNO returns it with a comment; the resident fixes it; the FNO approves.
9. **Structured output.** The approved review produces an output suitable for maternal death review — structured like Form 4, and showing for each value whether it was reused, derived, or entered, and from where.
10. **A case without a labour record.** Show the same workflow for a woman who died before labour or was referred in after delivery.
11. **History.** Show who did what, and when, across both records.

How you get there is your decision. We are not specifying screens, how the labour record is entered, how reuse is presented, or how the review is structured. We are interested in why you made the choices you did.

Requirements: essential, desirable, deferred

Complete the essential list well before reaching for anything else. Tell us which desirable items you chose and why, and what you deliberately left out.

Essential

1. **Overlap analysis.** A written mapping of every Form 4 item to the four categories in "The core question", with reasons and with the differences in terminology or precision you noticed.
2. **Labour records.** Synthetic labour records structured like the LCG, with timestamps and the author of each entry. **You do not need to build a full LCG charting interface.** Seeded data plus a simple way to add or view entries is enough; spend your time on reuse, not on drawing the chart.
3. **Review linked to the source.** A death review can be created from a woman's labour record — or without one.
4. **Proposed reuse with provenance.** Values the labour record can supply are proposed, each showing its source entry, time and original author. Derived values show how they were calculated.
5. **Clinician acceptance.** Each proposed value is accepted, rejected or corrected by the clinician. The labour record itself is never changed by the review.
6. **The rest of Form 4.** Everything else can be captured through a guided, conditional experience that hides what does not apply and can be resumed later.
7. **No unsupported conclusions.** Cause of death, diagnoses and contributing factors are only ever chosen by a clinician. Relevant observations may be shown as evidence.
8. **Gaps and late entries visible.** Missing observations are shown as missing. Entries added to the labour record after the time of death are identifiable.
9. **Checks before submission.** Missing, impossible or inconsistent information is caught and explained in plain language.
10. **Review and approval.** At least author (resident) and approver (FNO) roles, with return-with-comments and approval; approved reviews are locked and later changes are attributed amendments.
11. **Structured output.** An output suitable for maternal death review, following Form 4's structure, with the origin of each value available.

12. **History.** Who created, changed, accepted, submitted, returned and approved what, and when.
13. **Synthetic cases.** At least four: a death after a charted labour; a caesarean after a period of charted labour; a death with no labour record (before labour or after an abortion); and a woman referred in after delivering elsewhere.

Desirable — choose deliberately

- Form 1 produced from the same review with no re-entry.
- A timeline view combining labour observations and later events, distinguishing contemporaneous from retrospective entries.
- Alert-threshold crossings from the labour record surfaced as evidence next to the relevant review items.
- A second labour-record format (a partograph or a hospital variant) mapped without changing the review model.
- Printable Form 4 layout suitable for a wet signature.
- Two hospitals in one deployment, each unable to see the other's records.
- Automated tests for mapping, derivation and validation rules; one-command local setup.

Deferred — please do not build

- A full graphical LCG with plotting and alert logic.
- Connections to any government or hospital system (MPCDSR portal, HMIS, ABDM, hospital information systems).
- Issuing death certificates.
- A committee module with scheduling and notifications.
- Offline synchronisation (discuss it in your documents if you think it matters).
- AI that writes clinical narratives or suggests causes of death.
- Production deployment or compliance certification.

Technical freedom and open-source preference

Complete technical freedom. **We strongly prefer established, actively maintained open-source technology and avoid closed source dependencies.** Include a short **dependency register**: each significant component, its licence, why you chose it, and its maintenance status.

If you use a health-data standard (for example HL7 FHIR resources for observations), explain what it buys you and what it costs in seven days. You are not required to use one.

Security and data integrity

Keep in mind integrity as well as confidentiality. Keep in mind the DPDP rules.

Data model expectations

Choose whatever database and modelling approach you think fits, and explain why.

Synthetic data and safety rules

Use only invented data. These rules are part of the assignment.

- **Plausible, not perfect.** Labour records should be clinically coherent — times in order, observations at sensible intervals, alerts followed by an assessment — and should include realistic imperfections such as a missed observation. They do not need medical review. If an AI tool generates them, check them yourself.
- **Variety.** Cover the cases listed under Essential requirement 13. A brought-dead case is a good test if you have time.